INCH-POUND
MIL-DTL-24643/36E
22 August 2002
SUPERSEDING
MIL-C-24643/36D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS3SWU
(INCLUDING VARIATION LS3SWUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

This cable can be manufactured in three variations: LS3SWU and LS3SWUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First -	Copper conductor, AWG Size No. 18 (7 by 0.0152 inch), uncoated.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.023 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black, one white, and one red conductor for each triad.
Fourth -	Three conductors cabled together with a nominal lay of 3.5 inches to form each triad.
Fifth -	Binder tape over each triad, applied at manufacturer's option, applied helically with overlap.
Sixth -	Braided shield of AWG No. 34 or No. 36 uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.

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Construction (watertight) (continued)

Seventh - Shield insulation of two polyester tapes, per NEMA FI 4, outer tape sealed, An alternate shield insulation may consist of one polyester tape, per NEMA FI 4, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape. Diameter over completed triad 0.240 inch nominal.

Eighth - The required number of triads (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with No. 1, from the center outward. Fillers shall used as necessary to form a firm, well rounded assembly.

Ninth - Binder tape, applied helically with overlap.

Construction variant for type LS3SWU

Tenth - Cross-linked polyolefin jacket. (see table I for thickness).

Construction variant for type LS3WSUS

Tenth - Braided shield of tin coated copper. Minimum coverage of 85 percent.

Eleventh - Spirally wrapped 0.002 inch thick polyethylene (polyester film) Separator tape with 25 percent minimum overlap.

Twelfth - Braided shield of tin coated copper. Minimum coverage of 85 percent.

Thirteenth - Binder tape applied helically with overlap.

Fourteenth - Cross-linked polyolefin jacket. (see table I for thickness).

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TABLE I. Details.

Military part no. 1/ M24643/36	Size 2/	Number of triads	Cable jacket thickness min avg (inch)	Diameter over shielded core (inch nom.)	Overall diameter (inches)			
					TYPE LS3SWU		TYPE LS3SWUS	
					min	max	min	max
-01UN	3	3	0.040	0.561	0.610	0.655	0.675	0.710
-02UN	7	7	0.040	0.759	0.880	0.940	0.940	1.000
-03UN	10	10	0.050	1.024	1.100	1.180	1.160	1.240
-04UN	14	14	0.050	1.080	1.200	1.280	1.260	1.340
-05UN	19	19	0.050	1.243	1.370	1.450	1.430	1.510
-06UN	24	24	0.050	1.345	1.640	1.760	1.700	1.820
-07UN	30	30	0.050	1.430	1.760	1.860	1.820	1.920
-08UN	37	37	0.060	1.560	1.890	1.940	1.950	2.050
-09UN	44	44	0.060	1.780	2.140	2.240	2.200	2.300

1 When double overall shield is required, see MIL-DTL-24643 for configuration of part number.
2/ Type LS3SWU and LS3SWUS.

EXAMINATION AND TESTS:

Basic electrical: Requirements:

- Conductor resistance - ohms/1000 feet at 25°C, maximum 7.06
- Voltage withstand - volts, root mean square, minimum
  - Conductor to conductor 2000
  - Shield to shield (Triad shields only) 500
  - Shield to shield (LS3SWUS overall shield only) 200
- Insulation resistance - Megohms -1000 feet, minimum
  - Conductor to shield 500
  - Component shield to shield 100
  - Shield to shield (LS3SWUS overall shield only) 100
- Jacket flaws (LS3SWUS only) ..... No failure
- Conductor and shield (LS3SWUS only) continuity ..... No failure

Group A:

- Visual and dimensional ..... No failure
- Hydrostatic (open end) - leakage at 300 lb/in^2, 2 hours, in^3, maximum 1.0
- Capacitance
  - Mutual capacitance - 1 kilohertz (kHz), pF/ft, maximum 30
  - Capacitance unbalance - percent maximum 10

Group B:

- Cross-linked proof test (percent, maximum)
  - Insulation 50

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Jacket (When tested at 200°C) 50
Tear - pounds per inch thickness, minimum (ASTM D 470) 35
Physicals (unaged)
Insulation
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

Group C:
Physicals (aged)
Insulation
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (jacket) - cycles, minimum 125
Cable sealant removability ..... No failure
Shield - conformance to material, construction and coverage ..... No failure
Surface transfer impedance (type LS3SWUS only)
Milliohms per meter, maximum 700
EMP response time - dB, minimum 60

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Requirements:
Aging and compatibility (cable) (125 ± 5°C.) ..... No failure
Abrasion resistance (jacket) - scrapes, minimum 75
Acid gas equivalent - percent, maximum
Jacket 2
Fillers 2

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Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3 through 19	1000
24 through 44	500

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-034)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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